MD. MASHRUP HASAN

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Portfolio | LinkedIn | GitHub | ResearchGate | Google Scholar

SUMMARY

Geospatial researcher working at the intersection of GIScience, GeoAI, and spatial data science, with research spanning flood and disaster risk, environmental health, climate vulnerability and migration, land-use dynamics, and vision-based geospatial analytics. My research integrates remote sensing, geoinformatics, and machine learning to investigate human-environment interactions across terrestrial and coastal systems. Through both academic and applied research, I develop scalable geospatial methods for environmental monitoring, hazard and health risk assessment, ecosystem analysis, and spatial decision support. My broader research interest lies in leveraging multi-source geospatial data and artificial intelligence to understand landscape change, quantify its drivers and impacts, and generate evidence that supports climate resilience, disaster risk reduction, and sustainable environmental management.

EDUCATION

- **M.S. in Geography and Environment (CGPA 3.86/4.00 with Distinction)** *2023 – 2026*
  Merit Position: 2nd out of 18 students
  - Department of Geography and Environment, Shahjalal University of Science & Technology (SUST), Sylhet, Bangladesh.
  - **Thesis:** *Flood-Induced Health Risk Assessment in Sunamganj: A Deep Learning-Enhanced GIScience Approach.*
  - **Relevant coursework:** Natural Hazards and Disaster Management; Advanced GIS and Remote Sensing; Climate Change, Government Policy and Action Strategy; Water Resource Management; Urban Environmental Management.
- **B.Sc. in Geography and Environment (CGPA 3.70/4.00)** *2019 – 2024*
  Merit Position: 2nd out of 45 students
  - Department of Geography and Environment, Shahjalal University of Science & Technology (SUST), Sylhet, Bangladesh.
  - **Research Project:** *Suitability Analysis for Flood-Adaptive Green Infrastructure: Building Urban Resilience Through Strategic Placement in the Chattogram City Corporation.*
  - **Relevant coursework:** GIS and Remote Sensing; Hydrology and Fluvial Morphology; Climatology; Environmental and Social Impact Assessment; Quantitative Techniques in Geography; Research Methods in Geography and Environment.

RESEARCH AND WORK EXPERIENCE

- **Job:** Center for Participatory Research and Development (CPRD) *Oct 2025 – Present*
  Research Officer *Dhaka, Bangladesh*
  - Conduct mixed-methods research (household surveys, FGDs, and KIIs integrated with climate and health data) on climate-induced health risks as non-economic loss and damage (NELD) among climate migrants in Dhaka's informal settlements and Indigenous Peoples in the Northwest Barind Tract (2026–2027), informing policy analysis of Bangladesh's NAP, H-NAP, NDC, and National Mechanism on Loss and Damage.
  - Completed donor-funded projects on climate justice advocacy for LDCs; climate migration and protection gaps; the layered conflict among climate migrants, Rohingya refugees, and host communities; and climate-induced soil salinity and women's reproductive health.
  - Produce spatial analyses, thematic maps, and 5+ policy reports and briefings for government bodies and international donors, and contribute evidence to CSO position papers and multi-stakeholder consultations for national and UNFCCC advocacy.

- **Job:** Betterskill.io (e-Learning Platform), Bangladesh *Feb 2026 – Present*
  Course Developer & Instructor *Bangladesh*
  - Design and produce a 10-hour recorded course on Kobo Toolbox and NVivo, equipping learners with end-to-end survey design, field data collection, and qualitative analysis workflows.
  - Develop comprehensive course materials including video lectures, slides, datasets, quizzes, and assignments aligned with industry and research standards to ensure practical learning outcomes.
  - Deliver responsive one-to-one learner support by troubleshooting software queries, clarifying methodological concepts, and resolving workflow challenges, ensuring every learner receives targeted guidance tailored to their individual research or professional context.
- **Job:** Last Line Web Ltd., Sylhet *July 2025 – Sep 2025*
  Product Trainer and Research Assistant *Sylhet, Bangladesh*
  - Designed and delivered hands-on training programs on data collection tools and survey methodologies, raising client proficiency scores by 40% and closing workflow knowledge gaps across user cohorts.
  - Built and managed systematic data collection and MEAL workflows across project cycles, achieving zero overdue submissions over the tenure.
  - Conducted targeted needs assessments and gap analyses, designing structured evaluation criteria to measure training effectiveness, contributing to a 41% improvement in client retention outcomes.
  - Developed standardized report and manual templates that cut production turnaround time by 23% while maintaining version control and professional consistency.
- **Project:** SUST Research Center *Nov 2023 – Jul 2025*
  Research Assistant *Sylhet, Bangladesh*
  - Conducted GIS-based flood modelling, LULC classification, and spatial risk analysis across 3 studies, interpreting satellite imagery and remote sensing composites to quantify human-environmental interactions and urban vulnerability, producing decision-relevant outputs that contributed directly to 4 peer-reviewed journal publications.
  - Collected and validated primary environmental and spatial datasets through fieldwork and remote sensing interpretation, directly strengthening the empirical accuracy of research findings and map-based outputs.
  - Improved the analytical rigour and presentation of 10+ geospatial research manuscripts, ensuring spatial methodologies, figures, and results met institutional and journal submission standards.
- **Job:** College Education Development Project (CEDP), Murari Chand College *Jun 2024*
  Trainer *Sylhet, Bangladesh*
  - Delivered six structured sessions on document processing, graphic design, and video editing to college teachers, building practical digital skills under the Ministry of Education's Basic ICT Training Program.
  - Developed training materials that supported participant learning and ensured sessions were accessible, well-paced, and outcome focused.

TECHNICAL SKILLS

- **GIS & Remote Sensing:** ArcGIS Pro, ArcGIS Desktop, Google Earth Engine.
- **Programming & Scripting:** Python (GeoPandas, Rasterio, Shapely, Scikit-learn), R, HTML, JavaScript, JupyterLab, GitHub.
- **UAV & Photogrammetry:** UAV flight operations, Aerial data acquisition, Image Preprocessing, Ortho mosaic generation, DSM and DEM generation, and UAV-based spatial analysis.
- **Machine Learning & Deep Learning:** SVM, Random Forest, CNN, ANN, YOLO.
- **Data & Survey Tools:** SPSS, NVivo, Kobo Toolbox.
- **Document & Report Production:** Microsoft 365, Google Workspace, LaTeX, Zotero,
- **Design & Multimedia:** Adobe Illustrator, Adobe Photoshop, WordPress, Adobe After Effects.
- **Audio Engineering:** Audacity, FL Studio.

JOURNAL PUBLICATIONS

1. **Hasan, M. M.**, Jaman, F., & Nayem, J. (2026). *Mapping housing vulnerability to hailstorms in Bangladesh: A decision-support approach for disaster risk reduction.* Progress in Disaster Science. https://doi.org/10.1016/j.pdisas.2026.100643
2. Mostan, N., Kabir, M. N., **Hasan, M. M.**, & Islam, M. M. (2026). *Understanding morphological and hydrological changes in the Dhaleshwari River Basin through remote sensing and GIS-based analysis.* International Journal of River Basin Management. https://doi.org/10.1080/15715124.2025.2610998
3. Islam, M., Jaman, F., **Hasan, M. M.**, & Laskor, A. H. (2025). *Impact of extreme rainfall and land use-land cover change on flood hazards in Sylhet District of Bangladesh.* Case Studies in the Environment. https://doi.org/10.1525/cse.2025.2711193
4. **Hasan, M. M.**, Nishat, N. A., Ahsan, S. M. R., & Nayem, J. (2025). *Sylhet walkways as urban windows: Unveiling their role in urban life and environment stewardship.* International Journal of Science and Development, 2(1), 88–96.https://doi.org/10.5281/zenodo.15555477

WORKING PROJECTS

1. Rahman, S., Hossain, M. S., & **Hasan, M. M.** *Spatiotemporal Prediction of Fish Catch in the Bay of Bengal Using Random Forest Regression and Multi-Source Oceanographic Predictors* (Manuscript in preparation)
2. Niam, S., **Hasan, M. M.**, & Uddin, M. M. *Vision-only re-identification of individual plants across repeated UAV passes.* (In revision, Computers and Electronics in Agriculture)
3. Shamsuddoha, M., Anowar, S. B., Rahman, S., Hossain, M. S., Islam, H. R., & **Hasan, M. M.** *Environmental salinity exposure and pelvic inflammatory disease risk among women in climate-vulnerable coastal Bangladesh.* (Manuscript in preparation)
4. Paul, S., **Hasan, M. M.**, Basak, R., & Islam, M. A. *Impact of urbanization on microclimatic suitability in the context of public health: A GIS and RS-based approach in Sylhet City.* (Manuscript in preparation)
5. Koley, N. J., Nayeem, J., Jain, F., **Hasan, M. M.**, & Jaman, F. *Loss and damage impact on climate-induced migration nexus: Experience from climate-vulnerable communities in north-eastern Bangladesh.* (Manuscript in preparation)

CONFERENCE AND SYMPOSIUM

1. **Hasan, M. M.**, Islam, H. M. R., & Al Shariar, M. (2025, June). *Unregulated gher expansion and agrarian transition in Shyamnagar Upazila, Bangladesh: Land use conflict prediction and evidence-based coastal zoning.* International Symposium on Coast, Climate and Community (ISCCC). [Poster Presentation]
2. Talukder, A. A., **Hasan, M. M.**, Nishat, N. A., & Basak, R. (2025, December). *Industrial land conversion and local economic growth: Causal evidence from Olipur, Habiganj.* 1st International Conference on Life and Earth Sciences for Sustainable Development (ICLESSD-2025). [Oral Presentation]
3. Hossain, M. A., Rahman, M. M., Huda, R. B., Sani, M. S. S., & **Hasan, M. M.** (2025, November). *Exploring vertical growth patterns and regulatory compliance of high-rise mixed-use buildings in Dhaka South City Corporation.* International Conference on Geosciences for Achieving Sustainable Development Goals (ICGSDG). [Oral Presentation]
4. Sharmin, I., Nishat, N. A., Jaman, F., Sultan, F., & **Hasan, M. M.** (2025, April). *Carbon storage and vegetation dynamics in Ratargul Swamp Forest: A GIS-remote sensing approach.* 8th International Conference on Engineering Research, Innovation & Education (ICERIE 2025). [Oral Presentation]
5. Talukder, A. A., **Hasan, M. M.**, Sultana, F., Roy, B., & Nishat, N. A. (2025, February). *Spatial dynamics of environment, socioeconomics, and public health in urban transport hubs: A case study in Sylhet City Corporation.* RURS 1st International Student Conference and Research Fair. [Oral Presentation]
6. **Hasan, M. M.**, & Ahmed, T. S. (2022, October). *Sustainable tourism in Bangladesh: Mapping strategies for SDGs.* ExCEEd 2022, Civil and Environmental Engineering Department, SUST. [Poster Presentation]

TECHNICAL PROJECTS

- **RF-CPUE Bay of Bengal** - Random Forest pipeline predicting fisheries CPUE from oceanographic data.
  github.com/MashrupHasan/RF-CPUE-BayOfBengal
  - Built a 5-predictor Random Forest regression pipeline (mixed-layer thickness, salinity, chlorophyll, SST, surface currents) to predict yearly Catch Per Unit Effort across the Bay of Bengal from GLORYS and MODIS raster data, with grouped cross-validation and a Kriged-CPUE variant for full spatial interpolation.
- **CabbageTrack** - Vision-only UAV plant re-identification pipeline.
  github.com/MashrupHasan/CabbageTrack
  - Developed a pipeline combining YOLOv9c-seg detection, phase-correlation visual odometry, and cross-pass image registration to re-identify individual plants across repeated drone passes without GNSS, re-identifying 98.9% of blind-sampled plants (90/91) on a held-out flight.
- **StockTrek** - Multi-account investment platform for DSE/CSE investors.
  github.com/MashrupHasan/Investment_Tracker
  - Built a portfolio platform for Bangladesh stock-market investors that consolidates multiple Beneficiary Owner (BO) accounts, with live DSE price sync, an allocation planner, risk and dividend analytics, and a phone-friendly layout.

EXTRA-CURRICULLAR ACTIVITIES

- **Vice President, GEE Society, Sylhet (Dec 2024 – Jan 2026)**
  Facebook: GEE Society

  Oversaw planning and execution of seminars, training sessions, and awareness programs on Geography and Environment; fostered collaborative learning through mentorship.
- **General Secretary, SHIKORH Cultural Organization, Sylhet (Feb 2023 – Aug 2024)**
  Facebook: SHIKORH

  Led operations for 120+ members; spearheaded large-scale cultural events; coordinated disaster-response charity initiatives.

AFFILIATIONS

- Member, Geography and Environment Department Alumni Network, SUST.
- General Member, SHIKORH, SUST.

PROFESSIONAL TRAININGS

- Environmental Management Systems Awareness (ISO 14001:2015),
  AJA Bangladesh Ltd (Certificate No. AJA-EMS-0525-005) (May 2025)
- Introduction to Programming with Python,
  Institute of Information and Communication Technology (IICT), SUST - under EDGE Project, Bangladesh Computer Council, ICT Division (May–Dec 2024)
- Application of ArcGIS Pro for Regional Planning,
  GEE Society, Shahjalal University of Science & Technology, Sylhet (Nov 2023)
- Research Techniques of Social & Applied Sciences,
  GEE Society, Shahjalal University of Science & Technology, Sylhet (Sep 2023)
- Fecal Sludge Management in Cities: An Element of City-Wide Inclusive Sanitation,
  Civil and Environmental Engineering Department, SUST (Oct 2022)
- Participatory Mapping with OpenStreetMap for Humanitarian and Disaster Preparedness,
  YouthMappers at SUST (Aug 2022)
- Coastal Disasters: Bangladesh Perspective,
  Disaster Research Training and Management Center, University of Dhaka (Jul 2022)

LANGUAGES

Bengali: Native language.
English: Professional Working Proficiency - IELTS Academic Band 7.0 (CEFR C1, Jul 2025).

AWARDS AND HONORS

- Best Oral Presentation Award [3rd], 1st International Conference on Life and Earth Sciences for Sustainable Development (ICLESSD-2025), Jagannath University, Dhaka, Bangladesh. Dec 2025
- National Science and Technology (NST) Fellowship, Ministry of Science and Technology (MoST), Bangladesh. 2025
- 2nd Runner Up, Painting Contest - National Environment Carnival 2023, organized by Youthpreneur Network. 2023
- University Merit Scholarship, Shahjalal University of Science and Technology, Bangladesh. 2019, 2020, 2021

REFERENCES

Dr. Md. Anowarul Islam
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Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh.
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Relationship: Project Supervisor (B.Sc.)

Md Shamsuddoha
Chief Executive
Center for Participatory Research and Development (CPRD),
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Email: doha@cprdbd.org
Relationship: Employer and Direct Supervisor